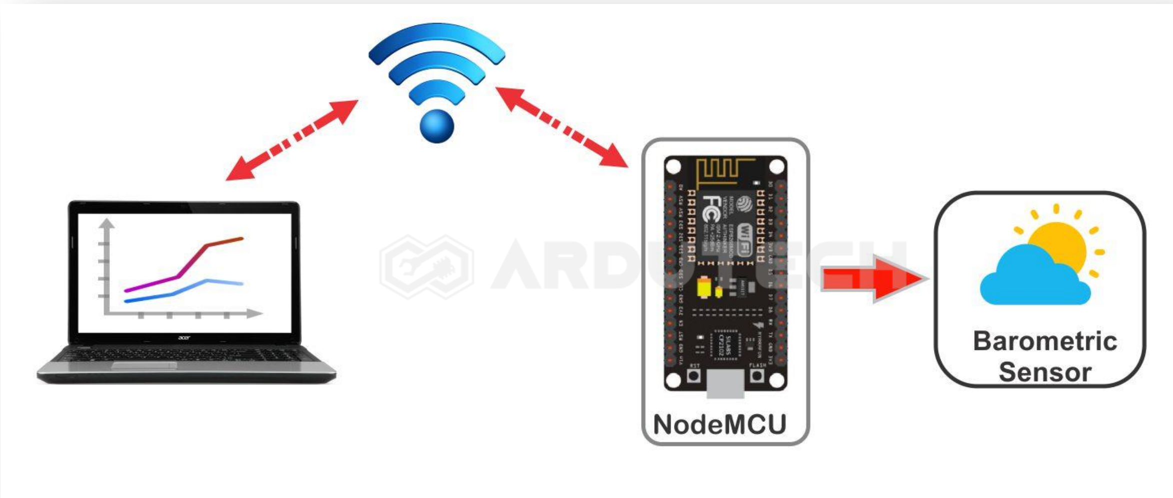
Membuat Weather Station dg Sensor BME280

Deskripsi

Membuat statiun pemantau cuaca yang memonitor temperatur, kelembaban, tekanan udara dan altitude dengan sensor BME280 dan hasilnya dikirim melalui jaringan WiFi dan ditampilkan dalam Web Page.

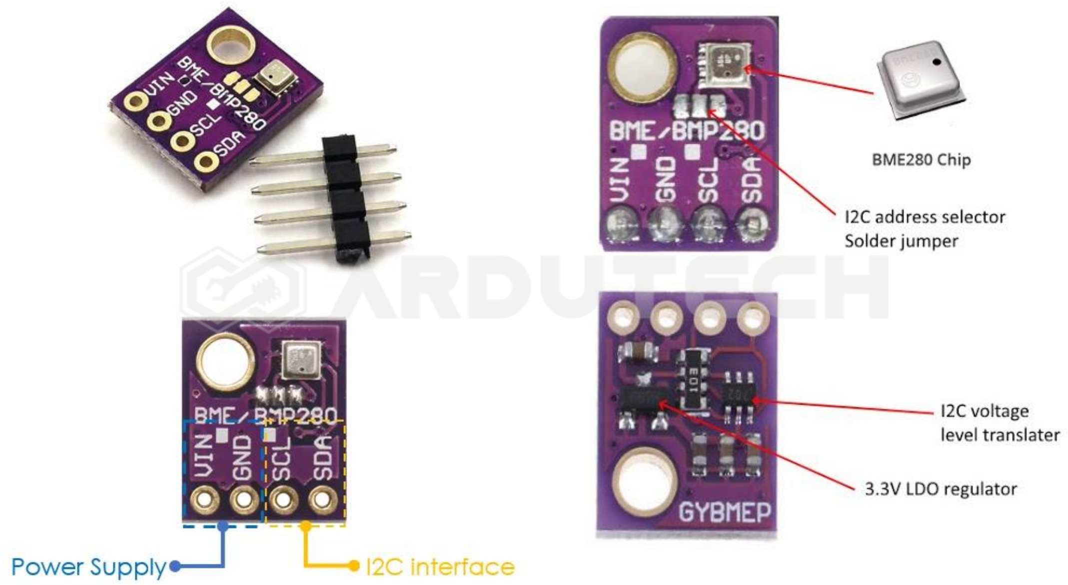
Cara Kerja

Sensor BME280 mengukur besaran temperature, kelembaban, tekanan udara dan altitude (ketinggian) kemudian mengirimkan data hasil pengukuran ke NodeMCU ESP8266. NodeMCU terhubung dengan jaringan internet (WiFi) kemudian mengirimkan datanya ke sebuah halaman web dengan alamat sesuai IP address-nya dan ditampilkan hasil pengukurannya.



BME280 merupakan sensor berukuran fisik kecil dengan kemampuan mengukur tekanan barometrik (berat udara), temperature, kelembaban dan juga ketinggian (altitude). Sensor ini dapat berkomunikasi dengan model I2C maupun SPI & pada proyek ini nanti kita akan memakai komunikasi I2C yang memanfaatkan kaki SDA dan SCL.

Sensor BMP280 sudah dikemas dalam sebuah modul yang terdiri dari 4 pin seperti terlihat pada gambar berikut ini.



Spesifikasi :

- Altimeter : Mengukur ketinggian tempat dari permukaan air laut (0 – 9,2KM) dg akurasi -+1M
- Thermometer : Mengukur suhu udara/ruangan (Range : -40 sampai +85 C)
- Humidity : mengukur kelembaban udara 0 – 100 %
- Atmospheric Pressure Meter : Mengukur tekanan udara disuatu tempat (Range : 300 hPa – 1100 hPa dg akurasi 12Pa)
- Komunikasi interface I2C dan SPI
- Address I2C : 0x76
- Akurasi relatif : 0.12hPa / 1 meter
- Akurasi absolut : 1hPa

Terdapat 4 pin koneksi pada sensor BME280 dengan keterangan sebagai berikut :

- VIN : tegangan input (3.3 – 5 V)
- GND : Ground
- SDA : pin data (komunikasi I2C)
- SDL : pin clock (komunikasi I2C)

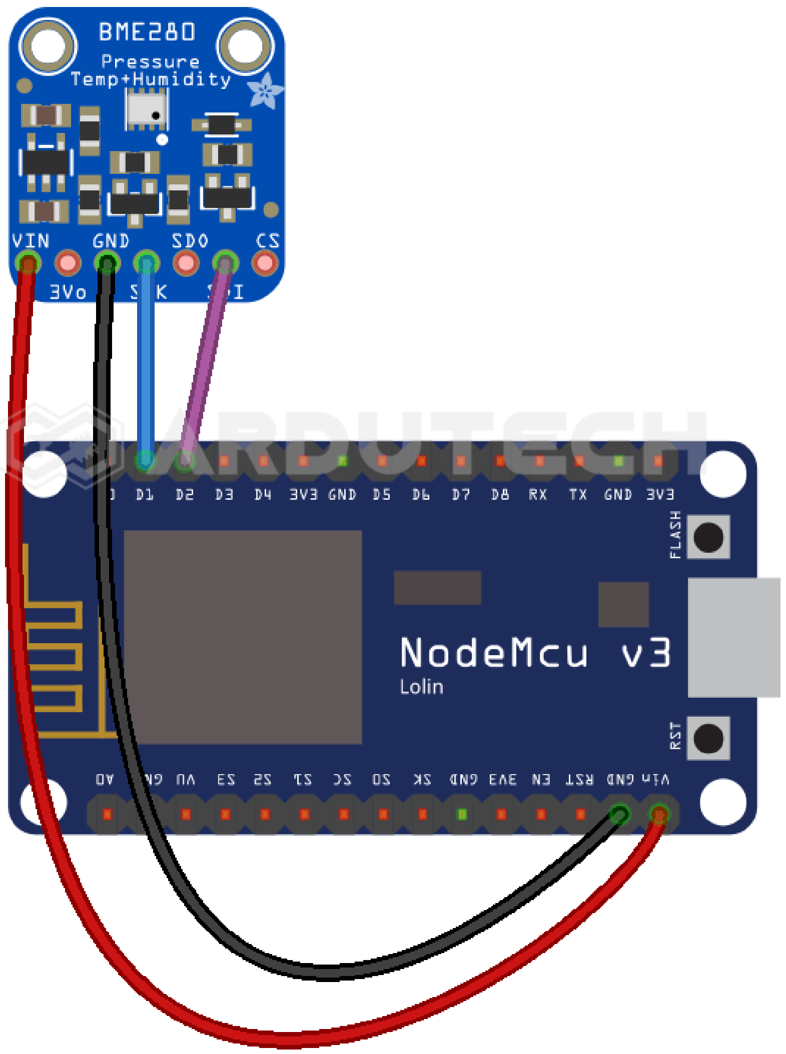
Kebutuhan Bahan

- NodeMCU V3
- Sensor Barometric Pressure BME280
- Kabel konektor
- Kabel micro USB

Kebutuhan Software

- Arduino IDE

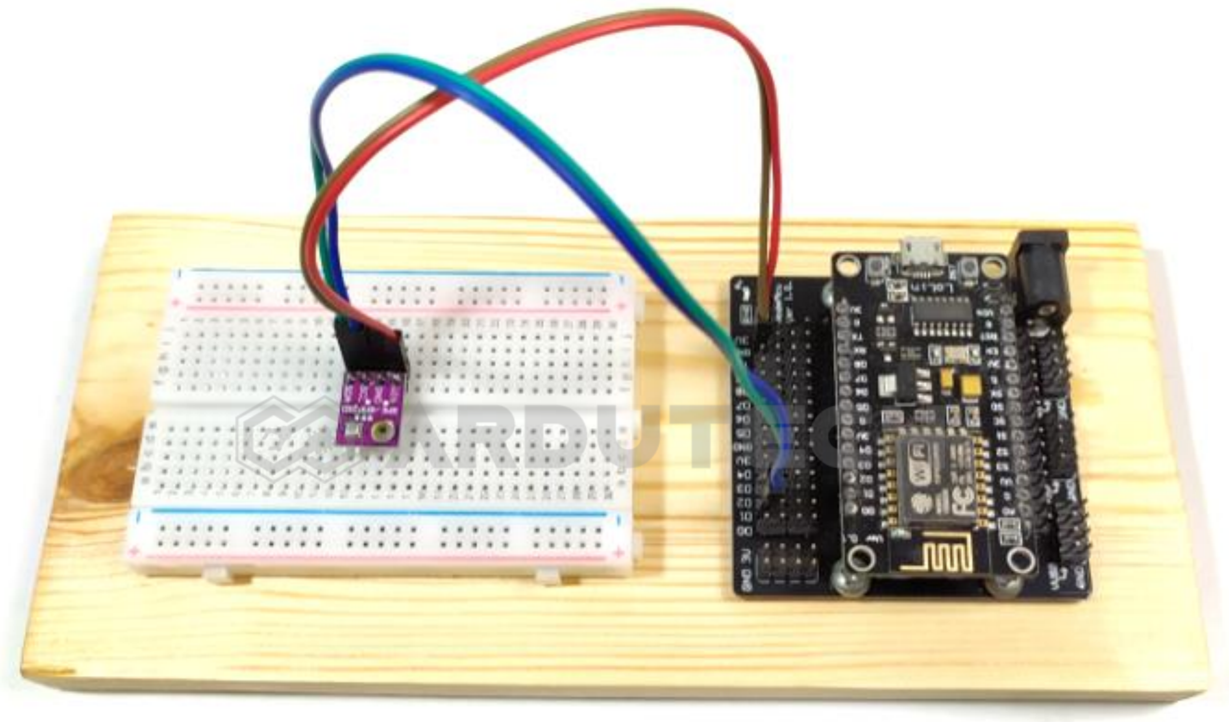
Rangkaian/Skematik



Koneksi NodeMCU dengan Sensor BME280 :

NodeMCU	BMP280
D1	SCL

D2	SDA
3V3	VCC
GND	GND



Program/Source Code

Program pada proyek ini memerlukan library :

- **ESP8266WiFi.h**
- **Wire.h**
- **Adafruit_Sensor.h**
- **Adafruit_BME280.h**

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/****************************************
 * Project Membuat Weather Station dg BME280
 * Board   : NodeMCU ESP8266 V3
 * Input   : BME280
 * Output : Web page
 * 99 Proyek IoT
 * www.ardutech.com
 ****************************************/
#include <ESP8266WebServer.h>
```

```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BME280.h>

#define SEALEVELPRESSURE_HPA (1013.25)
Adafruit_BME280 bme;
float temperature, humidity, pressure, altitude;
//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
const char* ssid = "ArdutechWiFi";  // Nama Hotspot/WiFi
const char* password = "12345678";  //  Password

ESP8266WebServer server(80);
//==================================
void setup() {
  Serial.begin(9600);
  delay(100);
  bme.begin(0x76);
  Serial.println("Connecting to ");
  Serial.println(ssid);
  //connect to your local wi-fi network
  WiFi.begin(ssid, password);
  //check wi-fi is connected to wi-fi network
  while (WiFi.status() != WL_CONNECTED) {
  delay(1000);
  Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFi connected..!");
  Serial.print("Got IP: ");  Serial.println(WiFi.localIP());
  server.on("/", handle_OnConnect);
  server.onNotFound(handle_NotFound);
```

```
  server.begin();
  Serial.println("HTTP server started");
}
//=========================================
void loop() {
  server.handleClient();
}
//=========================================
void handle_OnConnect() {
  temperature = bme.readTemperature();
  humidity = bme.readHumidity();
  pressure = bme.readPressure() / 100.0F;
  altitude = bme.readAltitude(SEALEVELPRESSURE_HPA);
  server.send(200,                                          "text/html",
SendHTML(temperature,humidity,pressure,altitude));
}
//========================================
void handle_NotFound(){
  server.send(404, "text/plain", "Not found");
}
//=======================================
String  SendHTML(float  temperature,float  humidity,float  pressure,float
altitude){
  String ptr = "<!DOCTYPE html>";
  ptr +="<html>";
  ptr +="<head>";
  ptr +="<title>Weather Station BME280</title>";
  ptr  +="<meta  name='viewport'  content='width=device-width,  initial-
scale=1.0'>";
  ptr                                                             +="<link
href='https://fonts.googleapis.com/css?family=Open+Sans:300,400,600'
rel='stylesheet'>";
  ptr +="<style>";
```

```
  ptr +="html { font-family: 'Open Sans', sans-serif; display: block; margin: 0px auto; text-align: center;color: #444444;}";
  ptr +="body{margin: 0px;} ";
  ptr +="h1 {margin: 50px auto 30px;} ";
  ptr +=".side-by-side{display: table-cell;vertical-align: middle;position: relative;}";
  ptr +=".text{font-weight: 600;font-size: 19px;width: 200px;}";
  ptr +=".reading{font-weight: 300;font-size: 50px;padding-right: 25px;}";
  ptr +=".temperature .reading{color: #F29C1F;}";
  ptr +=".humidity .reading{color: #3B97D3;}";
  ptr +=".pressure .reading{color: #26B99A;}";
  ptr +=".altitude .reading{color: #955BA5;}";
  ptr +=".superscript{font-size: 17px;font-weight: 600;position: absolute;top: 10px;}";
  ptr +=".data{padding: 10px;}";
  ptr +=".container{display: table;margin: 0 auto;}";
  ptr +=".icon{width:65px}";
  ptr +="</style>";
  ptr +="</head>";
  ptr +="<body>";
  ptr +="<h1>Weather Station BME280</h1>";
  ptr +="<div class='container'>";
  ptr +="<div class='data temperature'>";
  ptr +="<div class='side-by-side icon'>";
  ptr +="<svg enable-background='new 0 0 19.438 54.003'height=54.003px id=Layer_1 version=1.1 viewBox='0 0 19.438 54.003'width=19.438px x=0px xml:space=preserve xmlns=http://www.w3.org/2000/svg xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path d='M11.976,8.82v-2h4.084V6.063C16.06,2.715,13.345,0,9.996,0H9.313C5.965,0,3.252,2.715,3.252,6.063v30.982";
  ptr +="C1.261,38.825,0,41.403,0,44.286c0,5.367,4.351,9.718,9.719,9.718c5.368,0,9.719-4.351,9.719-9.718";
  ptr +="c0-2.943-1.312-5.574-3.378-7.355V18.436h-3.914v-2h3.914v-2.808h-4.084v-2h4.084V8.82H11.976z M15.302,44.833";
```

```
  ptr         +="c0,3.083-2.5,5.583-5.583,5.583s-5.583-2.5-5.583-5.583c0-
2.279,1.368-4.236,3.326-5.104V24.257C7.462,23.01,8.472,22,9.719,22";
  ptr
+="s2.257,1.01,2.257,2.257V39.73C13.934,40.597,15.302,42.554,15.302,44.83
3z'fill=#F29C21 /></g></svg>";
  ptr +="</div>";
  ptr +="<div class='side-by-side text'>Temperature</div>";
  ptr +="<div class='side-by-side reading'>";
  ptr +=(int)temperature;
  ptr +="<span class='superscript'>&deg;C</span></div>";
  ptr +="</div>";
  ptr +="<div class='data humidity'>";
  ptr +="<div class='side-by-side icon'>";
  ptr  +="<svg  enable-background='new  0  0  29.235  40.64'height=40.64px
id=Layer_1  version=1.1  viewBox='0  0  29.235  40.64'width=29.235px  x=0px
xml:space=preserve                       xmlns=http://www.w3.org/2000/svg
xmlns:xlink=http://www.w3.org/1999/xlink                      y=0px><path
d='M14.618,0C14.618,0,0,17.95,0,26.022C0,34.096,6.544,40.64,14.618,40.64s
14.617-6.544,14.617-14.617";
  ptr +="C29.235,17.95,14.618,0,14.618,0z M13.667,37.135c-5.604,0-10.162-
4.56-10.162-10.162c0-0.787,0.638-1.426,1.426-1.426";
  ptr
+="c0.787,0,1.425,0.639,1.425,1.426c0,4.031,3.28,7.312,7.311,7.312c0.787,
0,1.425,0.638,1.425,1.425";
  ptr         +="C15.093,36.497,14.455,37.135,13.667,37.135z'fill=#3C97D3
/></svg>";
  ptr +="</div>";
  ptr +="<div class='side-by-side text'>Humidity</div>";
  ptr +="<div class='side-by-side reading'>";
  ptr +=(int)humidity;
  ptr +="<span class='superscript'>%</span></div>";
  ptr +="</div>";
  ptr +="<div class='data pressure'>";
  ptr +="<div class='side-by-side icon'>";
  ptr  +="<svg  enable-background='new  0  0  40.542  40.541'height=40.541px
id=Layer_1  version=1.1  viewBox='0  0  40.542  40.541'width=40.542px  x=0px
xml:space=preserve                       xmlns=http://www.w3.org/2000/svg
```

```
xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path
d='M34.313,20.271c0-0.552,0.447-1,1-1h5.178c-0.236-4.841-2.163-9.228-
5.214-12.593l-3.425,3.424";

  ptr +="c-0.195,0.195-0.451,0.293-0.707,0.293s-0.512-0.098-0.707-0.293c-
0.391-0.391-0.391-1.023,0-1.414l3.425-3.424";

  ptr +="c-3.375-3.059-7.776-4.987-12.634-
5.215c0.015,0.067,0.041,0.13,0.041,0.202v4.687c0,0.552-0.447,1-1,1s-1-
0.448-1-1V0.25";

  ptr +="c0-0.071,0.026-0.134,0.041-
0.202C14.39,0.279,9.936,2.256,6.544,5.385l3.576,3.577c0.391,0.391,0.391,1
.024,0,1.414";

  ptr +="c-0.195,0.195-0.451,0.293-0.707,0.293s-0.512-0.098-0.707-
0.293L5.142,6.812c-2.98,3.348-4.858,7.682-5.092,12.459h4.804";

  ptr +="c0.552,0,1,0.448,1,1s-0.448,1-
1,1H0.05c0.525,10.728,9.362,19.271,20.22,19.271c10.857,0,19.696-
8.543,20.22-19.271h-5.178";

  ptr +="C34.76,21.271,34.313,20.823,34.313,20.271z M23.084,22.037c-
0.559,1.561-2.274,2.372-3.833,1.814";

  ptr +="c-1.561-0.557-2.373-2.272-1.815-3.833c0.372-1.041,1.263-
1.737,2.277-1.928L25.2,7.202L22.497,19.05";

  ptr +="C23.196,19.843,23.464,20.973,23.084,22.037z'fill=#26B999
/></g></svg>";

  ptr +="</div>";

  ptr +="<div class='side-by-side text'>Pressure</div>";

  ptr +="<div class='side-by-side reading'>";

  ptr +=(int)pressure;

  ptr +="<span class='superscript'>hPa</span></div>";

  ptr +="</div>";

  ptr +="<div class='data altitude'>";

  ptr +="<div class='side-by-side icon'>";

  ptr +="<svg enable-background='new 0 0 58.422 40.639'height=40.639px
id=Layer_1 version=1.1 viewBox='0 0 58.422 40.639'width=58.422px x=0px
xml:space=preserve xmlns=http://www.w3.org/2000/svg
xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path
d='M58.203,37.754l0.007-0.004L42.09,9.935l-0.001,0.001c-0.356-0.543-
0.969-0.902-1.667-0.902";

  ptr +="c-0.655,0-1.231,0.32-1.595,0.808l-0.011-0.007l-0.039,0.067c-
0.021,0.03-0.035,0.063-0.054,0.094L22.78,37.692l0.008,0.004";
```

```
  ptr                                       +="c-0.149,0.28-0.242,0.594-0.242,0.934c0,1.102,0.894,1.995,1.994,1.995v0.015h31.888c1.101,0,1.994-0.893,1.994-1.994";
  ptr +="C58.422,38.323,58.339,38.024,58.203,37.754z'fill=#955BA5 /><path d='M19.704,38.674l-0.013-0.004l13.544-23.522L25.13,1.156l-0.002,0.001C24.671,0.459,23.885,0,22.985,0";
  ptr        +="c-0.84,0-1.582,0.41-2.051,1.038l-0.016-0.01L20.87,1.114c-0.025,0.039-0.046,0.082-0.068,0.124L0.299,36.851l0.013,0.004";
  ptr +="C0.117,37.215,0,37.62,0,38.059c0,1.412,1.147,2.565,2.565,2.565v0.015h16.989c-0.091-0.256-0.149-0.526-0.149-0.813";
  ptr         +="C19.405,39.407,19.518,39.019,19.704,38.674z'fill=#955BA5 /></g></svg>";
  ptr +="</div>";
  ptr +="<div class='side-by-side text'>Altitude</div>";
  ptr +="<div class='side-by-side reading'>";
  ptr +=(int)altitude;
  ptr +="<span class='superscript'>m</span></div>";
  ptr +="</div>";
  ptr +="</div>";
  ptr +="</body>";
  ptr +="</html>";
  return ptr;
}
```

PERHATIKAN !

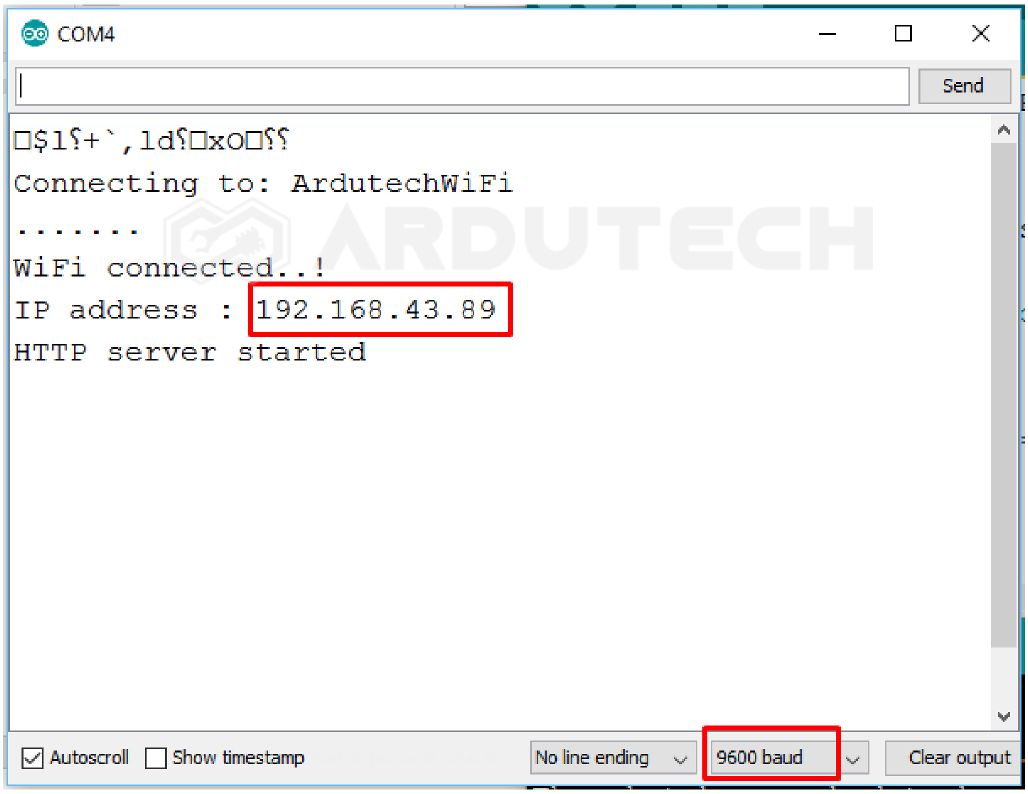
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **ssid [ ]**
- Password jaringan WiFi/hotspot : **password [ ]**

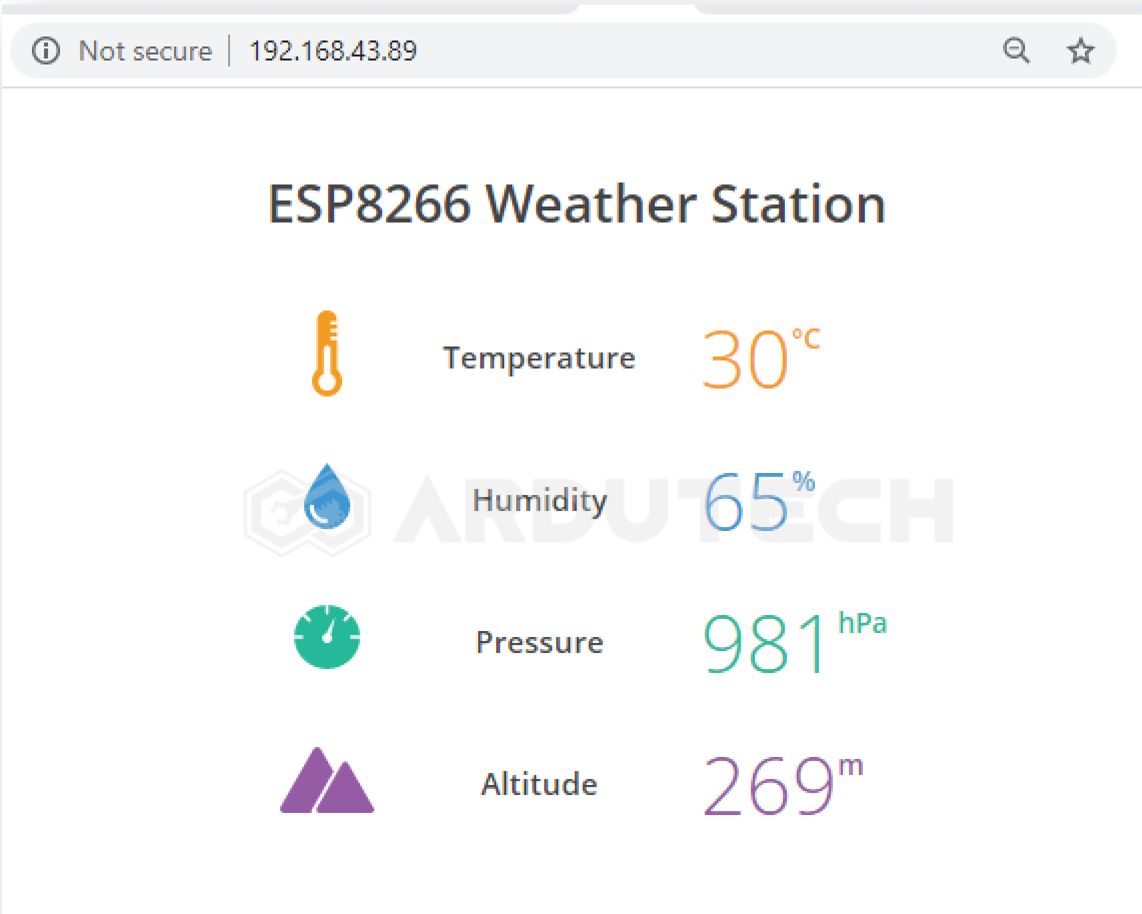
Simpan (**Save**) programnya kemudian Upload. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

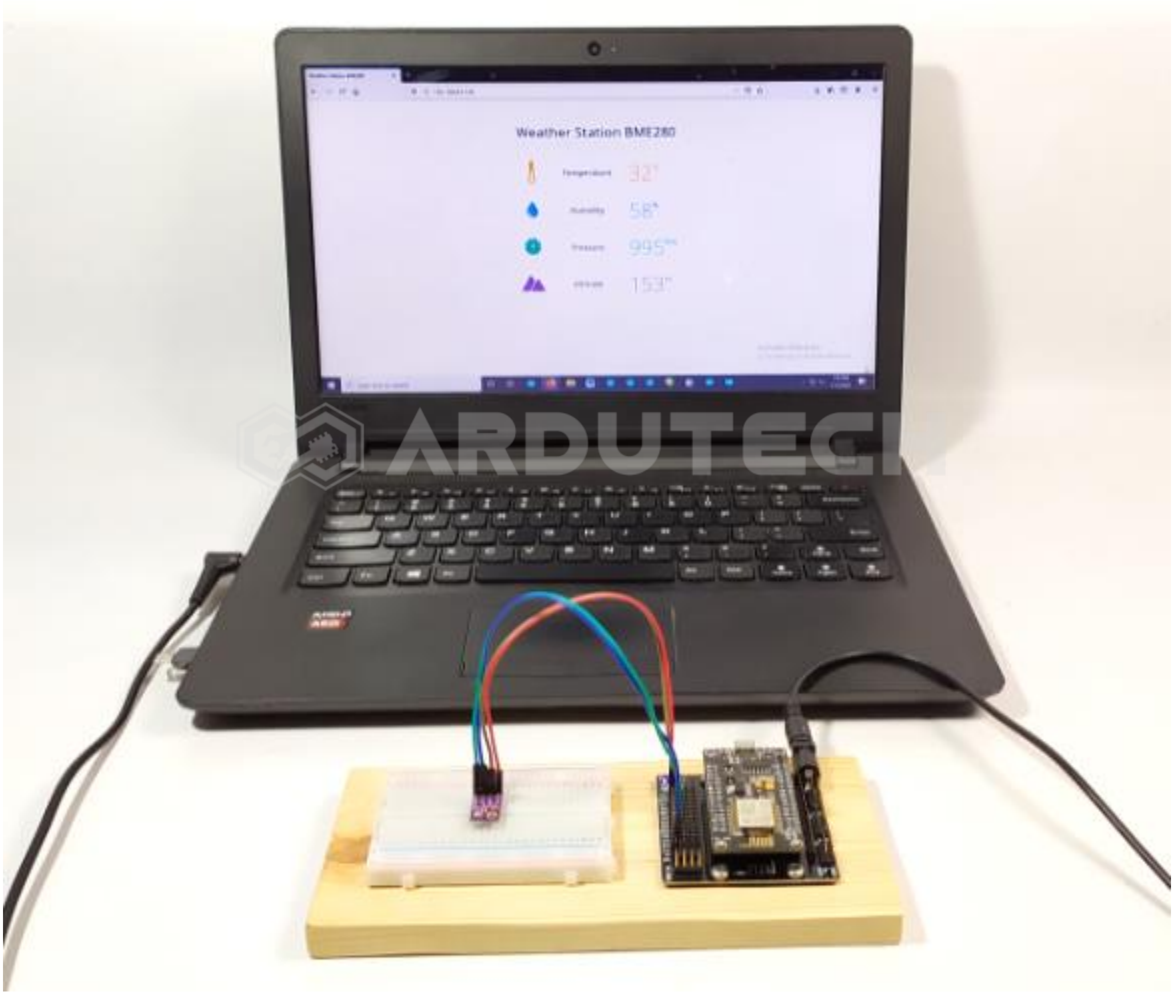
Jalannya Alat

Setelah program berhasil di Upload, silakan buka Serial Monitor dari menu Tools → Serial Monitor, seting baudrate pada 9600 :



Terlihat IP address pada Serial Monitor, sekarang buka di browser anda dan tuliskan IP address tadi :





Cobalah untuk memberi perubahan suhu pada sensor BME280 dan amati perubahannya.

Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – *"Sahabat Inovasi Anda"*